

HW# 2

Deep Learning for Computer Vision (EN525.733.8VL)

Due date: Feb. 12th, 2026

Q.1: Figure below shows a neural network involving a single hidden neuron for solving XOR problem. Show this network solves the XOR problem by constructing (a) decision regions for neuron O_1 (hidden neuron), (b) neuron O_2 (output neuron which has **three inputs decision boundary is a plane use MATLAB to plot**) and (c) a truth table for the network. **Activation function is the Step function for all the neurons.**

Input	Output
00	0
01	1
10	1
11	0

